Not All Pairs Need All Correspondences: Certified Adaptive Budgets for Robust Point Cloud Registration

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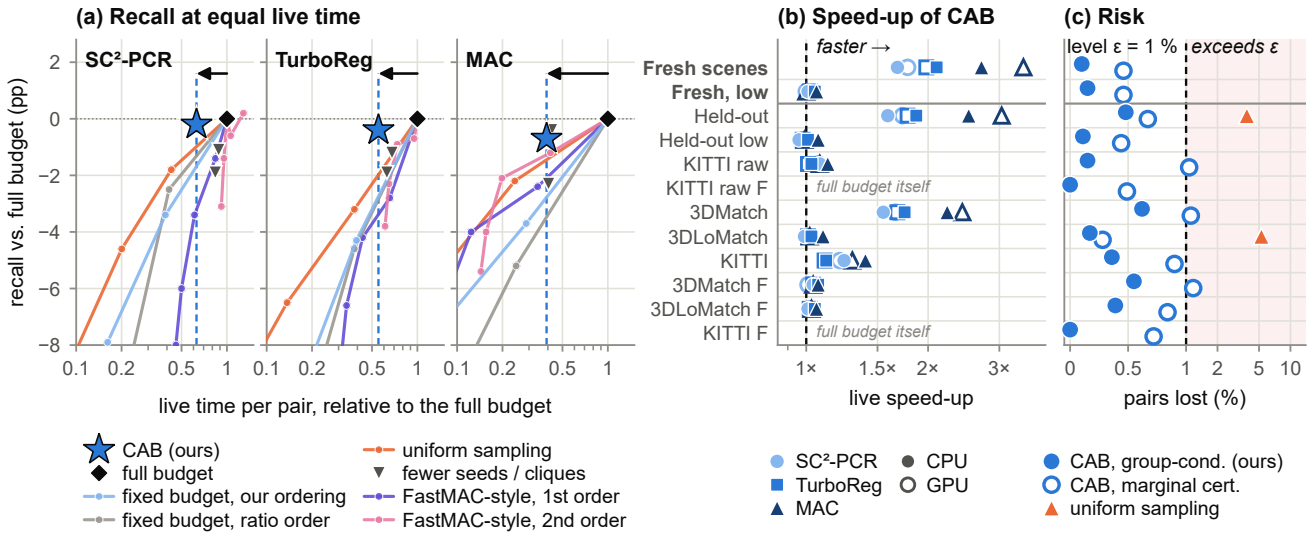


Figure 1. **Cheaper is easy; safe is not.** (a) Recall against live time per pair, both relative to the full budget on the same pairs, on the held-out scenes of protocol v2 (overlap above 30 %), for every cheaper way of running each estimator (legend, Tab. 3); CAB (star) adapts the budget to each pair, and the arrow marks its saving. (b) Live speed-up of CAB over the full budget on every test set (bold: blind; F: FCGF), for each estimator on a CPU (filled) and a GPU (open); with FCGF on KITTI the certified method is the full budget itself. Its recall is non-inferior in every case (Tabs. 1 and 2). (c) Risk: the share of pairs registered by the full estimator but lost, for each method at its worst over estimators and orderings. Only CAB with group-conditional certificates (filled blue) stays below the certified level of 1 % everywhere; marginal certificates (open) and uniform sampling of 2,000 correspondences exceed it on some test sets.

Abstract

The most accurate robust estimators for point cloud registration build a compatibility graph over all putative correspondences and pay this cost on every pair, although most pairs can be registered from a small, well-chosen subset. Making them cheaper is easy; making them cheaper safely *is not: every fixed shortcut, such as fewer correspondences, sampling or fewer seeds, is tuned to one benchmark's mix of easy and hard pairs and silently loses pairs when the mix changes. We present certified adaptive budgets (CAB), a drop-in wrapper that runs an unchanged estimator on growing budgets of mutually consistent correspondences and stops as soon as a learned controller is confident; a cheap gate sends pairs that would need the full budget straight to it. The stopping rule is certified per difficulty group with Learn-then-Test, so that with high probability at most 1 % of the pairs registered by the full estimator are lost, for any mix of difficulties. We evaluate it under pre-registered protocols with blind test sets, with hand-crafted and learned descriptors; the final method is tested once on 2,623 indoor pairs that no earlier experiment had processed. Around SC^2-PCR, TurboReg and MAC, CAB is 1.68×, 2.10× and 2.71× faster on these pairs with statistically non-inferior recall (1.78×, 1.98× and 3.44× on a GPU), and its realized risk stays within the certified level on all 72 test conditions, whereas marginal certificates and uniform sampling exceed it. Where nearly every pair needs the full budget, CAB declines to cut instead of losing pairs, at a cost of at most 4 % in time.*

1. Introduction

Correspondence-based registration aligns two scans from putative point correspondences proposed by local descriptors. The robust estimator that turns them into a rigid motion is where most of the accuracy comes from, and increasingly where most of the time goes. The most accurate estimators today, such as SC^2-PCR [6], MAC [32] and TurboReg [29], build a compatibility graph over all N correspondences, at a cost that grows quadratically or cubically with N. They spend this cost uniformly: every configuration fixes N for every pair, and evaluations sweep it from 250 to 5,000 [14, 22].

Yet pairs differ enormously in difficulty. Run on only 250 correspondences chosen by our ordering (Sec. 3.2), SC^2-PCR already registers 70.1 % of the pairs of our held-out indoor scenes in 29 ms, against 86.3 % in 660 ms with all 4,000 (live time). Other pairs need every correspondence, and some cannot be registered at all.

Cheaper is easy; safe is not. The obvious remedy is a cheaper fixed configuration, and Fig. 1(a) shows that one is easily found: uniform sampling of 2,000 correspondences runs SC^2-PCR in 282 instead of 660 ms on the held-out scenes. But a fixed shortcut is tuned to one benchmark's mix of easy and hard pairs, and nothing tells its user what it loses: 3.35 % of the pairs the full estimator registers on the held-out scenes and 4.83 % on low-overlap 3DLoMatch (Fig. 1(c)). A speed-up that drops an unknown share of the pairs the trusted estimator would have registered is not one that practitioners can adopt. What they need is a *contract*: a bound on what is lost relative to the estimator they trust, which survives a change of data.

Certified adaptive budgets. We provide such a contract with *certified adaptive budgets* (CAB), a drop-in wrapper around an unchanged estimator (Algorithm 1). CAB runs the estimator on a ladder of growing budgets and lets a learned controller stop after any stage. Three ingredients make this both fast and safe. First, the budgets are filled by a *compatibility-guided ordering* that puts mutually consistent correspondences first, scoring every correspondence against a small set of inlier-rich anchors at a cost far below that of the estimator's own graph (Fig. 2). Second, a cheap *hardness gate* recognises, before the estimator runs, pairs on which the ladder would only add cost, and runs the full budget on them directly; it can only remove losses. Third, the controller's operating point is certified with Learn-then-Test [2]: with probability $1 - \delta$, at most a fraction ε of the pairs that the full estimator registers are lost, and this holds *separately for every difficulty group*, hence for any mix of easy and hard pairs. The second point matters in practice: marginal certificates, valid only for the mix of difficulties they were calibrated on, exceed their level when the test mix differs (up to 1.17 % on our test sets), while difficulty-conditional ones hold on every test set.

Our contributions are:

- A drop-in wrapper that adapts the correspondence budget of a graph-based estimator to each pair, with a certified bound on the pairs lost against the full estimator that is valid for any mixture of difficulties, and a pre-estimation gate that keeps the bound (Sec. 3).
- A two-pass anchor ordering that fills small budgets with consistent correspondences at a fraction of the cost of the full graph, and an analysis of which pairs it can rescue (Secs. 3.2 and 4.5).
- Evidence under pre-registered protocols with blind test sets (held-out 3DMatch scenes, KITTI raw drives outside the odometry benchmark and, for the final method, pairs never processed before), FPFH and FCGF descriptors, three graph-based estimators, live timing on a CPU and a GPU and paired tests, that the certified wrapper stays within its risk level on every test set, while marginal certificates and fixed shortcuts (fewer seeds or cliques, uniform or FastMAC-style sampling, fixed budgets) do not (Secs. 4.2 to 4.4).

On the fresh blind pairs CAB is 1.68× faster for SC^2-PCR, 2.10× for TurboReg and 2.71× for MAC, the costliest of the three; where the data leave nothing to save, at low overlap and on outdoor raw drives, it declines to cut rather than lose pairs, and where its controller cannot tell success from failure, it runs the full budget.

2. Related work

Graph-based robust estimators. Spatial compatibility between correspondence pairs [3] is the backbone of today's most accurate robust estimators. SC^2-PCR [6] scores pairs by second-order compatibility and grows consensus sets from spectral seeds, MAC [32] enumerates maximal cliques, and TurboReg [29] replaces clique enumeration by pivot-guided search of 3-cliques; TEASER [30] and KISS-Matcher [17] prune outliers with certifiable or graph-theoretic tests. All of them build a graph over the N putative correspondences, so their cost is set by N, which each configuration fixes for every pair. FastMAC [33] reduces the number of correspondences passed to MAC by stochastic spectral sampling, but computes its sampling signal on the full second-order graph and uses one sampling ratio for all pairs; TurboReg notes that downsampling correspondences speeds up clique search at a cost in accuracy [29]. Among registration pipelines, DIPR [1] stops iterations early with hand-set thresholds, and BUFFER-X-Lite [16] skips feature scales when an inlier count exceeds a user-defined threshold, losing up to 1.6 pp of success on 3DMatch. CAB is complementary: it leaves the estimator unchanged, wraps any of them, and decides per pair how many correspondences it needs, with a bound on the accuracy this costs.

Sampling and stopping in RANSAC. RANSAC [10] adapts its number of hypotheses to the estimated inlier ratio; PROSAC [8] draws from quality-ordered correspondences, sequential tests [18] and preemption [19] discard poor hypotheses early, MAGSAC++ [5] and SupeRANSAC [4] improve scoring and sampling, BANSAC [20] adapts sampling online, and Schönberger *et al.* [25] show that the classical stopping criterion is miscalibrated. None of these rules bounds the accuracy lost against a reference estimator; Supp. I compares a certified adaptive pose search with RANSAC.

Adaptive computation with guarantees. Early exits and adaptive computation time [12, 27] let networks spend less on easy inputs, and CALM [26] and Fast-yet-Safe [15] certify early exits with Learn-then-Test [2], a distribution-free risk-control procedure. We bring this guarantee to robust geometric estimators, where the reference is the full estimator rather than a network's last layer, and show that its marginal form breaks under the shifts in difficulty that separate registration benchmarks. Monitoring anytime algorithms by dynamic programming [13] and the value of computation [23] motivate our Markov controller, and our difficulty-conditional certificates follow the group-conditional (Mondrian) guarantees of conformal prediction [28].

3. Certified adaptive budgets

3.1. Problem setup

A pair of scans comes with N putative correspondences $\mathcal{C} = \{(\mathbf{p}_i, \mathbf{q}_i)\}_{i=1}^{N}$ from descriptor matching, sorted by distinctiveness (Lowe ratio). A robust estimator E maps any subset of $\mathcal{C}$ to a rigid transformation; the standard configuration runs it once on all N correspondences, the *full budget*, which is our *reference*. CAB runs E along a ladder of budgets $N_1 < \cdots < N_K = N$: stage k returns $\hat{T}_k = E(\mathcal{C}_{\pi,N_k})$, the estimate from the first N_k correspondences of an ordering π, at cumulative cost C_k. Let $s_k \in \{0, 1\}$ indicate that $\hat{T}_k$ is correct. A stopping policy observes signals z_k after each stage and stops at τ, returning $\hat{T}_\tau$. We want the cheapest policy whose *risk*, the fraction of pairs the reference registers but the policy does not, is at most ε:

$$\min_{\tau} \mathbb{E}[C_\tau] \quad \text{s.t.} \quad R(\tau) = \Pr\left(s_K = 1,\ s_\tau = 0\right) \le \varepsilon. \tag{1}$$

R upper-bounds the recall lost against the standard estimator. Measuring the risk against the reference rather than against ground truth isolates the price of adaptivity: CAB does not claim to fix the estimator's failures, only to bound how many of its successes it gives up, which is what a practitioner who has chosen an estimator cares about. Ground truth is needed only on calibration pairs, never at test time. Two things decide how cheap a certified policy can be: how often the early stages already succeed, which depends on the ordering π (Sec. 3.2), and how well the controller recognises those successes (Secs. 3.3 and 3.4).

3.2. Compatibility-guided ordering

Distinctiveness is a poor filter for a small budget: on the 3DMatch validation scenes, only 27.8 of the 250 most distinctive correspondences are inliers on average. Inliers, however, are mutually consistent. Rigidity implies $\left|\|\mathbf{p}_i - \mathbf{p}_j\| - \|\mathbf{q}_i - \mathbf{q}_j\|\right| < d$ for two inliers i, j, where d is the estimator's compatibility threshold; let C_{ij} indicate this test. Scoring all pairs (i, j) costs $O(N^2)$, as much as the estimator itself. We instead score every correspondence against a small set A of anchors with second-order support,

$$s_i = \sum_{a \in A} C_{ia} \left(C\, C_{AA}\right)_{ia}, \tag{2}$$

the number of anchor triangles that i closes, and fill each budget with the highest-scoring correspondences (ties keep the ratio order). A budget that covers all correspondences keeps the standard input, so the full stage is exactly the reference.

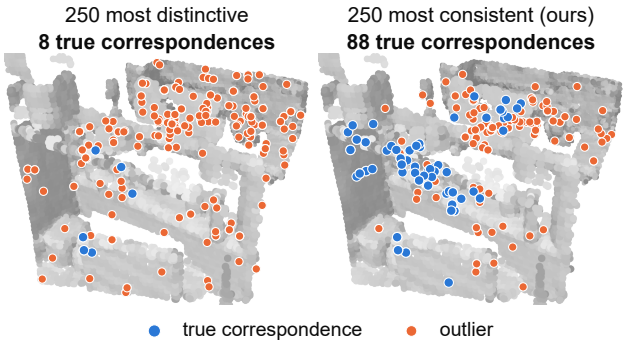


Figure 2. **What the cheap budget contains.** The 250 correspondences given to the estimator on a 3DMatch test pair, drawn at their reference endpoints on the reference fragment, seen from slightly above its sensor. Left: the 250 most distinctive (Lowe ratio). Right: the 250 most consistent with inlier-rich anchors (ours). Blue: true correspondences; orange: outliers.

Anchors decide. The obvious anchors, the $m = 100$ most distinctive correspondences, fail exactly where help is needed: in hard pairs they are nearly all outliers (12.5 inliers on average and none in 15 % of validation pairs), and outliers that happen to agree with each other are promoted. We therefore select anchors in a first pass: among the $P = 500$ most distinctive correspondences, the $m = 100$ with the highest first-order degree $\sum_{j \le P} C_{ij}$ become anchors, which raises the share of inliers among anchors to 35.6 %. The two passes cost $O(P^2 + Nm^2)$ instead of the $O(N^2)$ to $O(N^3)$ of the estimators' graphs, 7.2 ms per pair on one CPU core (Fig. 2).

3.3. Stopping controller

After each stage the controller observes estimator-agnostic signals: the estimator's inlier count; the fraction of *all* N correspondences, and of the 250 most distinctive, explained by $\hat{T}_k$; the change of $\hat{T}_k$ since the previous stage; the gain in explained correspondences; the stage index, $\log N$ and the median Lowe ratio of the budget. Two classifiers fitted on the fitting pairs predict $p_{\text{now}} = \Pr(s_k{=}1 \mid z_k)$ and the value of continuing, $p_{\text{fin}} = \Pr(s_K{=}1 \mid z_k)$; their kind (logistic regression with two regularisation strengths, quadratic logistic regression or gradient-boosted trees) is chosen with the ladder on the fitting pairs (Sec. 3.4). We consider three policy families: a *threshold* rule that stops when $p_{\text{now}} \geq \lambda$; *accept-or-give-up*, which also stops when $p_{\text{fin}} < \gamma$; and a *Markov* controller, a finite-horizon decision process over (stage, quintile of p_{now}, tercile of p_{fin}) whose transitions and costs are estimated on the fitting pairs' trajectories and which is solved by backward induction [21] for the reward $s - \lambda C$. The ladder itself is a design choice: candidates are increasing subsets of $\{250, 500, 1000, 2000, 4000\}$, including ladders that end below the full budget.

Hardness gate. Some pairs pay for every stage of the ladder and end at the full budget anyway. Before the estimator runs, the gate computes cheap signals of the correspondences, the first pass of the ordering (anchor degrees), the anchors' second-order support of all correspondences and Lowe-ratio statistics, and a classifier predicts whether CAB would cost more than the full budget on the pair; if so, the full budget runs directly. The gated result is the reference itself, so the gate can only remove losses and needs no certificate of its own: every guarantee below holds for the gated policy unchanged. Its classifier and threshold minimise the out-of-fold total cost on the fitting and calibration pairs.

Algorithm 1 CAB at test time

Require: correspondences $\mathcal{C}$ (ratio order), estimator E, ladder $N_1 < \cdots < N_K$, models p_{now}, group edges, certified thresholds λ_g

π, $h \leftarrow$ two-pass compatibility order of $\mathcal{C}$ (Eq. (2)) and gate signals
if gate(h) **then return** $E(\mathcal{C})$ ▷ the full budget, directly
end if
for $k = 1, \ldots, K$ **do**
  $\hat{T}_k \leftarrow E(\text{first } N_k \text{ of } \pi(\mathcal{C}))$ ▷ all of $\mathcal{C}$, in ratio order, if $N_k \geq |\mathcal{C}|$
  $z_k \leftarrow$ signals of $\hat{T}_k$ on $\mathcal{C}$ and change from $\hat{T}_{k-1}$
  if $k = 1$ **then** $g \leftarrow$ difficulty group of z_1
  end if
  if $p_{\text{now}}(z_k) \geq \lambda_g$ **or** $k = K$ **then return** $\hat{T}_k$
  end if
end for

3.4. Certification

Every policy family is indexed by a parameter ordered from conservative to aggressive. On n held-out calibration pairs, Learn-then-Test [2] tests $H_\lambda : R(\lambda) > \varepsilon$ in this fixed order with exact binomial p-values and keeps the most aggressive λ before the first non-rejection; then $\Pr\left(R(\hat{\lambda}) \leq \varepsilon\right) \geq 1 - \delta$ for test pairs exchangeable with the calibration pairs. We use $\varepsilon = 1\,\%$ and $\delta = 0.1$. A certificate needs data: even with no loss among n calibration pairs, the test rejects only if $(1-\varepsilon)^n \leq \delta$, *i.e.* for $n \geq 230$ here.

Difficulty-conditional certificates. Benchmark splits differ mostly in their mix of easy and hard pairs, which breaks exchangeability between calibration and test pairs. We therefore partition pairs into G groups by an observable after the first stage, the number of explained correspondences, at its terciles on the fitting pairs, and certify one threshold λ_g per group, each on that group's calibration pairs at level δ/G (which needs at least 339 pairs per group for $G = 3$).

Proposition 1. *If, within every group g, test pairs are exchangeable with that group's calibration pairs, then with probability at least $1 - \delta$ the policy with thresholds $(\hat{\lambda}_g)_g$ has $R \leq \varepsilon$ on every test distribution, whatever its group proportions.*

Proof sketch. By a union bound over the G group-wise guarantees, $R_g(\hat{\lambda}_g) \leq \varepsilon$ holds for all g simultaneously with probability $\geq 1-\delta$; then $R = \sum_g w_g R_g \leq \varepsilon$ for any proportions w_g (full proof in Supp. A). □

A marginal certificate only covers test sets with the calibration mix of difficulties; the conditional one covers any mix, at the price of a smaller δ per group and hence more calibration pairs.

Choose first, then certify. The guarantee covers one sequence of tests fixed in advance, so the ladder and the policy family are chosen before any calibration pair is used. On the fitting pairs, scored out of fold by controllers fitted on the other folds, every candidate (ladder, policy family and classifier kind) gets the most aggressive parameters whose losses stay within what the certificate can accept for the calibration set's size (none in any group, with our pool sizes), and we keep the cheapest candidate; only this one is certified, and the result is evaluated once on test. Certifying several candidates on the same calibration pairs and keeping the cheapest would void the guarantee, because the cheapest is the most likely to have been certified by chance. A ladder that stops below the full budget is risky even when it never stops early, so its always-continue rule is the first element of its sequence, and if that rule is rejected CAB falls back to the full budget. The full budget alone is always a candidate and needs no certificate: when nothing cheaper can be certified, for lack of calibration data or of a reliable controller, CAB is the standard estimator.

4. Experiments

4.1. Setup

Protocol. All data splits, rules and configurations were pre-registered before the data were downloaded or processed, in two stages (code release, Supp. B). Protocol v2 evaluated CAB without the gate, with the logistic controller, once on blind held-out scenes and KITTI raw drives. The gate and the classifier choice were then found on development data only, so protocol v3 tested the final method once on a fresh blind test: 2,623 pairs of 10 further 3DMatch training scenes that no experiment had processed before, under any role (377 of the 3,000 sampled pairs had appeared in earlier logs as fitting or training pairs and were removed before any was processed). The final method was fitted on fitting pairs outside these scenes. The protocol-v2 blind sets and the official benchmarks, which earlier versions of this work had evaluated, are reported for the final method as earlier test sets, not as blind tests.

Indoor data. The 75 3DMatch [31] training scenes are permuted with a fixed seed: 9 scenes form the blind test set, 13 provide 3,000 calibration pairs, and the rest 1,000 fitting pairs; pairs overlap by more than 10 %. The blind pairs are reported in two parts, above 30 % overlap (2,092 pairs, 3DMatch-like) and between 10 and 30 % (908, 3DLoMatch-like); the fresh blind test of protocol v3 likewise (1,976 and 647 pairs). Development benchmarks: the official 3DMatch (1,623) and 3DLoMatch [14] (1,781) test pairs, with the same fitting and calibration pairs.

Outdoor data. The 1,310 blind pairs come from the 49 KITTI [11] raw drives that belong to none of the odometry sequences 0–10, paired by FCGF's rule [7] with GPS/IMU poses refined by ICP; 22 pairs on which the refinement failed were dropped by a pre-registered rule. 300 odometry training pairs fit the controller and 1,238 other odometry training and validation pairs certify it; the 555 odometry test pairs are the development benchmark.

Correspondences and estimators. FPFH [24] descriptors (5 cm voxels indoors, 30 cm outdoors); for every source point the nearest target descriptor, sorted by Lowe ratio and capped at $N = 4{,}000$. FCGF [7] features are computed for every split with the official weights by our re-implementation of the network, which reproduces the released test features up to rounding (Supp. B). FCGF was trained on all 3DMatch training scenes and on odometry sequences 0–5, so it has no blind indoor test; it is fitted on the indoor fitting pairs, certified on the 1,331 pairs of the 3DMatch validation scenes and tested on the development benchmarks, and outdoors it is certified on 1,107 pairs of 42 raw drives and tested blind on the 203 pairs of the other 7. SC^2-PCR [6], TurboReg [29] and MAC [32], added by pre-registered addenda before it ran on any test pair, are re-implemented on CPU, MAC with its authors' reduction of dense graphs. At the full budget all three come within 0.5–1.3 pp of the published recall on 3DMatch and 3DLoMatch with FCGF and within 2.0–4.4 pp with FPFH (Supp. B).

Metrics. A pair is registered if RE $\leq 15^\circ$ and TE ≤ 30 cm indoors, and RE $\leq 5^\circ$ and TE ≤ 60 cm outdoors; recall (RR) is over all pairs. The *loss* of a configuration is the recall lost against the full budget with its one-sided 95 % upper bound (scene-stratified bootstrap [9]); it is *non-inferior* if the bound is below 1 pp (2 pp at low overlap), and its *risk* is the share of pairs registered by the full budget but lost.

Cost. *Wall-clock speed-ups* are measured *live*: one process with the default multi-threaded BLAS on a laptop CPU (AMD Ryzen 7 5700U, 8 cores), the same 300 random test pairs per set for every configuration, and the order of the runs rotated from pair to pair. Each run includes everything the configuration needs, such as the ordering in CAB or the sampling signal in FastMAC. The live timing is repeated on a GPU (NVIDIA T4) with PyTorch ports of the estimators and of CAB (Supp. E). Because time depends on the machine, we also count *operations*: the dominant arithmetic of every stage a pair visits (compatibility tests, graph products, consensus and hypothesis scoring, $O(N^2)$ to $O(N^3)$ in the budget N) plus the $O(P^2 + Nm^2)$ of our ordering, averaged over the full test set (Supp. B). Live and offline stopping decisions agree on at least 94 % of the pairs in every live run and 99.7 % indoors; the differences come from multi-threaded arithmetic, which on KITTI also changes a few nearest neighbours.

4.2. Main results

Table 1 reports the fresh blind test, with every choice made on the fitting pairs and every certificate computed on the calibration pairs before any blind pair was processed. On pairs above 30 % overlap, CAB runs SC^2-PCR 1.68× and TurboReg 2.10× faster than the standard full budget (1.96× and 2.45× fewer operations) and loses 0.00 and 0.00 pp of recall, with one-sided 95 % bounds of 0.00 and 0.00 pp: both are statistically non-inferior, and the realized risks of 0.00 and 0.00 % are inside the certified 1 %. At low overlap the gate sends most pairs to the full budget directly, and CAB is 1.01× and 1.05× faster, no longer slower than the full budget. On every earlier test set (Tab. 2) CAB stays within the risk level and is non-inferior, and on the KITTI raw drives the gate removes the slowdown of protocol v2 (1.08× and 1.03×, against 0.93× and 0.91×). MAC, whose full budget is the costliest of the three (4,033 ms per fresh pair above 30 % overlap, against 616 and 958 ms), gains most (Fig. 1(b)): CAB runs it 2.71× faster above 30 % overlap and 1.06× at low overlap, at realized risks of 0.10 and 0.00 %, and 1.40× on KITTI odometry. On a GPU the speed-ups follow those on the CPU on all 10 test sets (Supp. E): 1.78×, 1.98× and 3.44× above 30 % overlap on the fresh pairs.

Table 1. **Fresh blind test** (protocol v3: pairs never processed before, evaluated once; compatibility order, group-conditional certificates). RR: recall (%). Loss: recall lost against the full budget (pp) with its one-sided 95 % upper bound; ✓: non-inferior (bound below 1 pp; 2 pp at low overlap). Time: live wall-clock speed-up over the full budget on 300 random pairs per set, every configuration timed on the same pairs. Ops: reduction of the hardware-independent operation count over the full test set. Risk: pairs registered by the full budget but lost (%, target ≤ 1). CAB without gate (v2): the configuration of protocol v2 (logistic controller, no gate), fitted on the same pairs. MAC: pre-registered addenda.

	Fresh scenes, overlap >30 % (1,976 pairs)					Fresh scenes, overlap 10–30 % (647)				
Method	RR	Loss [UB]	Time	Ops	Risk	RR	Loss [UB]	Time	Ops	Risk
SC^2-PCR										
Full budget (standard)	93.1	–	1.00×	1.00×	–	52.9	–	1.00×	1.00×	–
CAB without gate (v2)	93.1	0.00 [0.00] ✓	1.78×	1.95×	0.00	52.9	0.00 [0.00] ✓	0.98×	1.04×	0.00
CAB (ours)	**93.1**	0.00 [0.00] ✓	**1.68**×	**1.96**×	0.00	**52.9**	0.00 [0.00] ✓	**1.01**×	**1.05**×	0.00
TurboReg										
Full budget (standard)	92.9	–	1.00×	1.00×	–	51.8	–	1.00×	1.00×	–
CAB without gate (v2)	92.8	0.10 [0.20] ✓	1.64×	1.70×	0.10	51.8	0.00 [0.00] ✓	1.00×	1.04×	0.00
CAB (ours)	**92.9**	0.00 [0.00] ✓	**2.10**×	**2.45**×	0.00	**51.6**	0.15 [0.46] ✓	**1.05**×	**1.10**×	0.15
MAC										
Full budget (standard)	92.6	–	1.00×	1.00×	–	54.3	–	1.00×	1.00×	–
CAB without gate (v2)	92.5	0.10 [0.20] ✓	2.56×	2.03×	0.10	54.3	0.00 [0.00] ✓	1.06×	1.06×	0.00
CAB (ours)	**92.5**	0.10 [0.25] ✓	**2.71**×	**2.42**×	0.10	**54.3**	0.00 [0.00] ✓	**1.06**×	**1.10**×	0.00

Table 2. **Earlier test sets** (protocol-v2 blind sets and development benchmarks; not blind for the final method). Live speed-up of CAB without gate (v2) and of CAB (*offline; †full budget, Sec. 4.3); CAB's operation reduction and risk (%). CAB is non-inferior in every case unless marked ‡. Complete results: Supp. F.

	SC^2-PCR				TurboReg				MAC			
Test set	v2	CAB	Ops	Risk	v2	CAB	Ops	Risk	v2	CAB	Ops	Risk
Blind test sets of protocol v2												
Held-out scenes	1.56×	**1.59**×	1.83×	0.24	1.49×	**1.87**×	2.09×	0.38	1.70×	**2.52**×	2.19×	0.48
Held-out scenes, low overlap	0.94×	**0.96**×	1.04×	0.11	0.95×	**1.01**×	1.08×	0.00	1.01×	**1.07**×	1.08×	0.11
KITTI raw	0.93×	**1.08**×	1.18×	0.15	0.91×	**1.03**×	1.13×	0.00	1.13×	**1.13**×	1.23×	0.15
KITTI raw (FCGF)	1.00×†	**1.00**×†	1.00×	0.00	1.00×†	**1.00**×†	1.00×	0.00	1.00×†	**1.00**×†	1.00×	0.00
Development benchmarks												
3DMatch	1.58×	**1.55**×	1.75×	0.37	1.42×	**1.75**×	1.94×	0.55	1.62×	**2.23**×	1.96×	0.62
3DLoMatch	0.96×	**0.99**×	1.03×	0.11	0.97×	**1.03**×	1.08×	0.17	1.01×	**1.10**×	1.07×	0.17
KITTI	1.08×	**1.24**×	1.54×	0.36	1.07×	**1.12**×	1.35×	0.18	1.35×	**1.40**×	1.58×	0.18
3DMatch (FCGF)	1.04×	**1.04**×	1.04×	0.18	1.01×	**1.07**×	1.09×	0.55	1.11×	**1.07**×	1.07×	0.18
3DLoMatch (FCGF)	1.08×	**1.01**×	1.07×	0.11	1.03×	**1.05**×	1.17×	0.28	1.13×	**1.06**×	1.10×	0.11
KITTI (FCGF)	1.00×†	**1.00**×†	1.00×	0.00	1.00×†	**1.00**×†	1.00×	0.00	1.00×†	**1.00**×†	1.00×	0.00

4.3. The guarantee under shift

A certificate is only worth having if it holds on data that differ from the calibration data. Figure 1(c) collects the realized risk on every test set: CAB stays within the 1 % level in all 72 cases, MAC included (at most 0.62 %). Two alternatives do not, each for a different reason.

Marginal certificates hold only for the calibration mix of difficulties. With the final method they exceed their level on 3DMatch and on FCGF 3DMatch (1.11 and 1.17 % for SC^2-PCR), and in protocol v2 they did on the blind KITTI raw drives, whose pairs are much harder than the odometry calibration pairs (61.8 vs. 92.3 % full-budget recall): 1.37 % for TurboReg with the ratio order. With MAC they exceed it on the blind KITTI raw drives (1.07 %) and, without the gate, on the fresh low-overlap pairs (2.47 %). The group-conditional certificates stay within the level in every case, as Proposition 1 predicts. In protocol v2 the marginal certificates' risk within one difficulty group reached 1.67 %, against at most 0.67 % for the group-conditional ones (Supp. G).

Fixed shortcuts carry no certificate. Uniform sampling of 2,000 correspondences loses 3.35 and 3.82 % of the pairs registered by the full estimator on the held-out scenes, and 4.83 and 5.27 % on 3DLoMatch (Sec. 4.4).

Learned descriptors. With FCGF the guarantee holds again, but the savings are small (Tab. 2): the pairs FCGF solves are solved at the first stage, and its few failures are hard to tell from successes; on KITTI, whose fitting pairs are FCGF's own training pairs, the controller learns no failure signal, and the certificate keeps the full budget, although 250 correspondences register 92.1 % of the blind pairs.

4.4. Equal-time comparison

Is adaptivity needed, or would any cheaper way of running the estimator do as well? Figure 1(a) and Tab. 3 compare CAB with five alternatives on the same pairs and the same live clock, for all three estimators (MAC: protocol v3, addendum D). Every alternative that runs as fast as CAB registers fewer pairs at its time, and to match CAB's recall each needs 1.1–2.3 times CAB's time. *Fewer seeds or cliques* barely

Table 3. **Equal-time comparison** on the held-out scenes of protocol v2 (not blind for the final method). Each cheaper variant is a recall–time curve closed by the full budget (live time, recall on every second test pair, 1,046 pairs). Min.: cheapest configuration (ms). At CAB: the variant's recall at CAB's time (%), and CAB's advantage on the same pairs (pp). Match: time the variant needs to reach CAB's recall.

Variant	Min.	At CAB	Δ	Match
SC²-PCR: full budget 660 ms, 86.3 %; CAB 415 ms, 86.1 %				
Fewer seeds (ratio 0.025, 0.01)	553	–	–	632 (1.5×)
Uniform sampling	27	85.1	+1.0	618 (1.5×)
FastMAC-style, 1st-order graph	303	83.1	+3.0	645 (1.6×)
FastMAC-style, 2nd-order graph	606	–	–	656 (1.6×)
Fixed budget, ratio order	28	84.7	+1.4	629 (1.5×)
Fixed budget, compat. order	29	84.2	+1.9	637 (1.5×)
TurboReg: full budget 1,027 ms, 87.6 %; CAB 567 ms, 87.2 %				
Fewer cliques (K_1=500, 250)	644	–	–	917 (1.6×)
Uniform sampling	5	85.3	+1.9	948 (1.7×)
FastMAC-style, 1st-order graph	310	84.3	+2.9	972 (1.7×)
FastMAC-style, 2nd-order graph	627	–	–	967 (1.7×)
Fixed budget, ratio order	10	84.3	+2.9	972 (1.7×)
Fixed budget, compat. order	62	84.4	+2.8	969 (1.7×)
MAC: full budget 4,195 ms, 87.4 %; CAB 1,645 ms, 86.7 %				
Top-K cliques (K=200, 100)	1,695	–	–	1,757 (1.1×)
Uniform sampling	15	85.6	+1.1	3,182 (1.9×)
FastMAC-style, 1st-order graph	312	85.2	+1.5	3,390 (2.1×)
FastMAC-style, 2nd-order graph	604	88.2	+0.7	2,970 (1.8×)
Fixed budget, ratio order	18	83.2	+3.5	3,769 (2.3×)
Fixed budget, compat. order	43	84.2	+2.5	3,629 (2.2×)

save time for SC²-PCR and TurboReg (cheapest settings 553 and 644 ms, full budgets 660 and 1,027 ms), because building the compatibility graph over all correspondences dominates; MAC's own Top-K clique selection saves more (1,695 of 4,195 ms) and comes closest, but still needs 1.1× CAB's time to match its recall. *FastMAC-style sampling* as published pays for its signal on the full second-order graph (584 ms on a CPU); it is strongest for MAC, for which it was designed, yet at CAB's time it registers 0.7 pp fewer pairs. *Fixed budgets* reach any time but trail CAB by 1.4–1.9 pp (SC²-PCR), 2.8–2.9 pp (TurboReg) and 2.5–3.5 pp (MAC) at its time, because a fixed rule must serve the hardest pairs it keeps, while CAB adds correspondences only where the first budget failed. *Uniform sampling* registers 1.0, 1.9 and 1.1 pp fewer pairs at CAB's time; with $N = 2{,}000$ it is faster than CAB but registers fewer pairs (McNemar $p = 0.03$, 2.5×10^{-5} and 0.05). On 3DLoMatch (Supp. H) a net recall difference can hide losses: FastMAC-style sampling of 2,000 correspondences keeps MAC's recall (+0.0 pp at 0.79× the time) but loses 1.79 % of the pairs the full budget registers, and every cheaper variant that saves at least 5 % of the time loses at least 4.04, 2.69 and 1.35 % of them (SC²-PCR, TurboReg, MAC), above the certified 1 %; CAB runs almost every pair to the full budget (1.00, 0.99 and 0.95× the full time).

4.5. When it pays off, and why

Table 4. **When CAB pays off.** Per test set, blind sets first (SC²-PCR / TurboReg): recall of the cheapest budget ($N = 250$, compatibility order, %), share of pairs no stage registers (%), live speed-up of CAB ([†]full budget, see Sec. 4.3) and realized risk (%).

Test set	N=250	Unsolvable	Speed-up	Risk
Fresh scenes	76.5 / 76.6	7 / 7	1.68 / 2.10	0.00 / 0.00
Fresh scenes, low overlap	25.2 / 25.0	47 / 47	1.01 / 1.05	0.00 / 0.15
Held-out scenes	68.8 / 68.5	14 / 14	1.59 / 1.87	0.24 / 0.38
Held-out scenes, low overlap	23.5 / 23.6	58 / 58	0.96 / 1.01	0.11 / 0.00
KITTI raw	6.3 / 7.2	38 / 37	1.08 / 1.03	0.15 / 0.00
KITTI raw (FCGF)	92.1 / 92.1	7 / 8	1.00[†] / 1.00[†]	0.00 / 0.00
3DMatch	65.1 / 65.0	18 / 17	1.55 / 1.75	0.37 / 0.55
3DLoMatch	21.4 / 21.6	62 / 62	0.99 / 1.03	0.11 / 0.17
KITTI	12.8 / 14.2	8 / 7	1.24 / 1.12	0.36 / 0.18
3DMatch (FCGF)	86.8 / 87.0	6 / 7	1.04 / 1.07	0.18 / 0.55
3DLoMatch (FCGF)	44.1 / 43.6	40 / 40	1.01 / 1.05	0.11 / 0.28
KITTI (FCGF)	92.4 / 92.4	6 / 6	1.00[†] / 1.00[†]	0.00 / 0.00

When it pays off. Table 4 lines up all test sets. The speed-up is set mainly by the data, through how often a small budget already succeeds and how few pairs are unsolvable: with the same wrapper, it is largest where 250 correspondences register about two thirds to three quarters of the pairs (fresh and held-out scenes, 3DMatch), small on KITTI odometry, where most pairs need 2,000 correspondences (12.8 % at $N = 250$), and close to one at low overlap, where 47 % of the fresh pairs and 58 % of the held-out pairs are unsolvable and hard to recognise early (after the first stage, final success is predicted with an AUROC of only 0.68 on 3DLoMatch, against 0.82 on 3DMatch); on the raw drives, where only 6.3 % of the pairs are registered at $N = 250$, the gate turns the slowdown of protocol v2 into a small gain. The risk stays within the level in every row: where there is nothing to save, CAB spends the full budget rather than losing pairs, at the price of the first stage it tries on the pairs the gate lets through (lowest speed-up 0.96×, SC²-PCR on the held-out scenes at low overlap), which also lengthens the slowest pairs (Sec. 5).

Middle-difficulty pairs. The ordering changes what the cheapest budget can do: with $N = 250$, SC²-PCR registers 65.1 % of 3DMatch pairs instead of 50.8 %, and 12.8 % of KITTI pairs instead of 1.6 %. Figure 3(a) shows for which pairs. On 3DMatch and 3DLoMatch, the gain is 23.0 to 33.6 pp for pairs with 5–49 true anchors and below

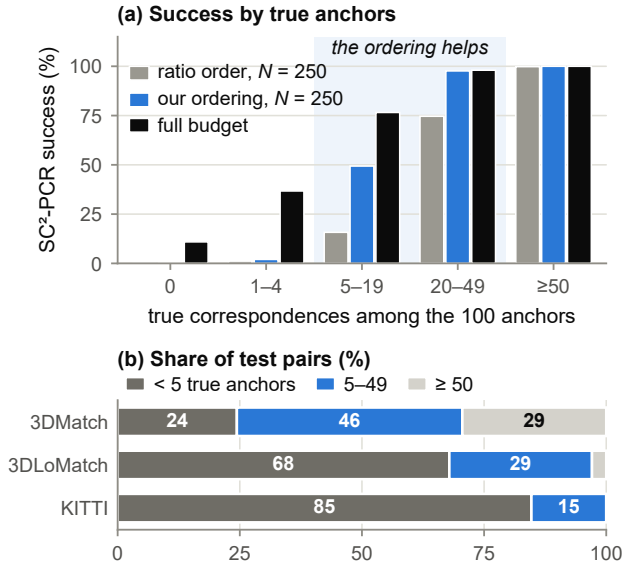


Figure 3. **Which pairs a cheap budget can rescue.** (a) SC^2-PCR success on the 3DMatch and 3DLoMatch test pairs, by the number of true correspondences among the 100 anchors: at $N = 250$, our ordering (blue) recovers most of the full-budget success for 5–49 true anchors (shaded) and nothing below. (b) Share of the test pairs with fewer than 5, 5–49 and at least 50 true anchors: only the middle group (blue) can be rescued by a cheap budget, and its share differs most between test sets.

1 pp elsewhere: there, pairs with fewer than five are rarely registered even with the full budget, and pairs with fifty or more were solved already. Figure 3(b) shows why the benefit varies across datasets: 68 % of 3DLoMatch pairs have fewer than five true anchors, against 24 % on 3DMatch. On KITTI 85 % do, yet the full budget registers 92.3 %: the inliers are rarely among the most distinctive correspondences, so these pairs need large budgets (Tab. 4).

4.6. Ablations

Anchors. Two passes make the ordering work (Supp. Tab. 17): one pass with the most distinctive correspondences as anchors barely changes SC^2-PCR's success at $N = 250$, two passes gain +7.5 pp.

Certificates. Marginal certificates are faster (2.31× against 1.67× for SC^2-PCR on the held-out scenes, offline) but exceed their level elsewhere (Sec. 4.3); the risk level and the number of groups trade speed for risk smoothly (Supp. G).

Gate and classifier. On the fresh pairs and the same clock, the final method is 1.28× [1.18, 1.39] as fast as CAB without the gate (v2) for TurboReg and 1.06× for MAC above 30 % overlap and at least as fast at low overlap, but 0.95× [0.92, 0.97] for SC^2-PCR above 30 % overlap, where the gate's live overhead outweighs its saving (Sec. 5); on the GPU, the final method is faster than CAB without the gate in 23 of the 30 timed conditions, SC^2-PCR above 30 % overlap included (1.78× against 1.74×). Added one at a time on the earlier test sets (Supp. D, offline), the gate is faster on 16 of 16 conditions where CAB cuts and never riskier, as it only sends pairs to the full budget; choosing the classifier kind is faster on 10 and slower on 5, and raises the risk on 11, by at most 0.49 pp, within the level. Combined, the final method is faster on 14 and slower on 2.

5. Limitations and conclusion

Limitations. Our estimators are re-implementations; on a GPU our PyTorch ports keep the savings (Supp. E), but they are not tuned for speed, and in an optimised GPU implementation fixed per-run costs could narrow the gap between small and large budgets. CAB lowers the mean time per pair, not the worst case: a pair that needs the whole ladder also pays for the stages tried before, up to 32 % more than one full run at the 99th percentile (TurboReg on KITTI odometry). Live, the gate costs 40–48 ms per pair (median over pairs with identical decisions), far more than the 3.2 ms of its prediction timed alone, for a reason we could not establish (Supp. C); it makes CAB slower than without the gate for SC^2-PCR above 30 % overlap, and the blind results include it (on the GPU it does not, Sec. 4.6). Only the fresh test is blind for the final method, whose gate and classifier choice were found on development data after the protocol-v2 evaluation; it is indoor and FPFH only, and its scenes are fresh with respect to the data that fit, calibrate and choose the final method, but every 3DMatch training scene had appeared in earlier logs. A certificate needs calibration data (the group-conditional one appeared from about a thousand pairs on random subsets, Supp. G) and, to be fast, fitting data with failures; FCGF has seen every 3DMatch training scene, so learned descriptors are tested blind only outdoors.

Conclusion. Not every pair needs every correspondence, but a cheaper estimator is only useful if one knows what it loses. Consistency-ordered budgets, a difficulty-conditional certificate of a candidate chosen beforehand and a pre-estimation gate make graph-based estimators markedly faster where the data allow, at statistically non-inferior recall and within the certified risk on every test set, including pairs that no experiment had processed before; marginal certificates and fixed shortcuts give no such assurance. A certificate is only as fast as its controller, which needs fitting data with failures. The headroom left (Supp. Tab. 16) lies in correct poses that the controller does not yet recognise and in pairs that no budget can register: sharper verification of each stage's pose, rather than the choice of budget, is the next step.

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